Automated Multi-Parametric MRI Segmentation of Post-Treatment Glioma Sub-regions

using Classical Machine Learning and Active Contours

Project Report Week 3

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1. Objective of Week 3

The objective of Week 3 was to design and implement the core tumor segmentation pipeline using classical machine learning techniques. The goal was to process multi-parametric MRI scans from the BraTS dataset and generate voxel-wise tumor predictions using a CPU-based system.

2. Problem Space Analysis

Most modern medical image segmentation methods rely on deep learning architectures such as 3D U-Net or transformer-based networks. Although these models achieve high accuracy, they require powerful GPU hardware and extensive training time.

This project explores an alternative approach by investigating whether meaningful tumor segmentation can be achieved using classical machine learning methods combined with strong feature engineering.

3. General Approach

The proposed segmentation pipeline consists of the following stages:

- Brain masking
- Feature extraction
- Random Forest classification
- Post-processing
- Visualization and evaluation

3.1 Brain Masking

A brain mask was generated by combining the four MRI modalities. Voxels with non-zero intensity in any modality were considered valid brain tissue. This removes background air and reduces unnecessary computation.

3.2 Feature Engineering

Instead of increasing model complexity, the system focuses on extracting meaningful voxel-level features including:

- Multi-modal intensity values from T1n, T1c, T2w and FLAIR

- Normalized spatial coordinates
- Radial distance from brain center
- Gradient magnitude features
- Local neighborhood mean and variance using a $3 \times 3 \times 3$ filter

A total of 18 features are extracted per voxel.

3.3 Model Training

A Random Forest classifier was used to classify each voxel as tumor or background. Random Forests are robust ensemble models that perform well with high-dimensional feature spaces.

3.4 Post Processing

Because voxel predictions are independent, small false-positive regions may appear. Morphological operations and connected component analysis were applied to retain only the largest anatomically valid tumor region.

4. Implementation Pipeline

The project was implemented using a modular architecture consisting of:

- Data loading module
- Preprocessing module
- Feature extraction module
- Random Forest training module
- Inference module
- Evaluation module
- Visualization module

A central pipeline script orchestrates loading MRI scans, extracting features, training the model, generating predictions, and evaluating segmentation results.

5. Segmentation Results and Visualization

Segmentation results were visualized using FLAIR MRI slices along with ground truth and predicted tumor masks.

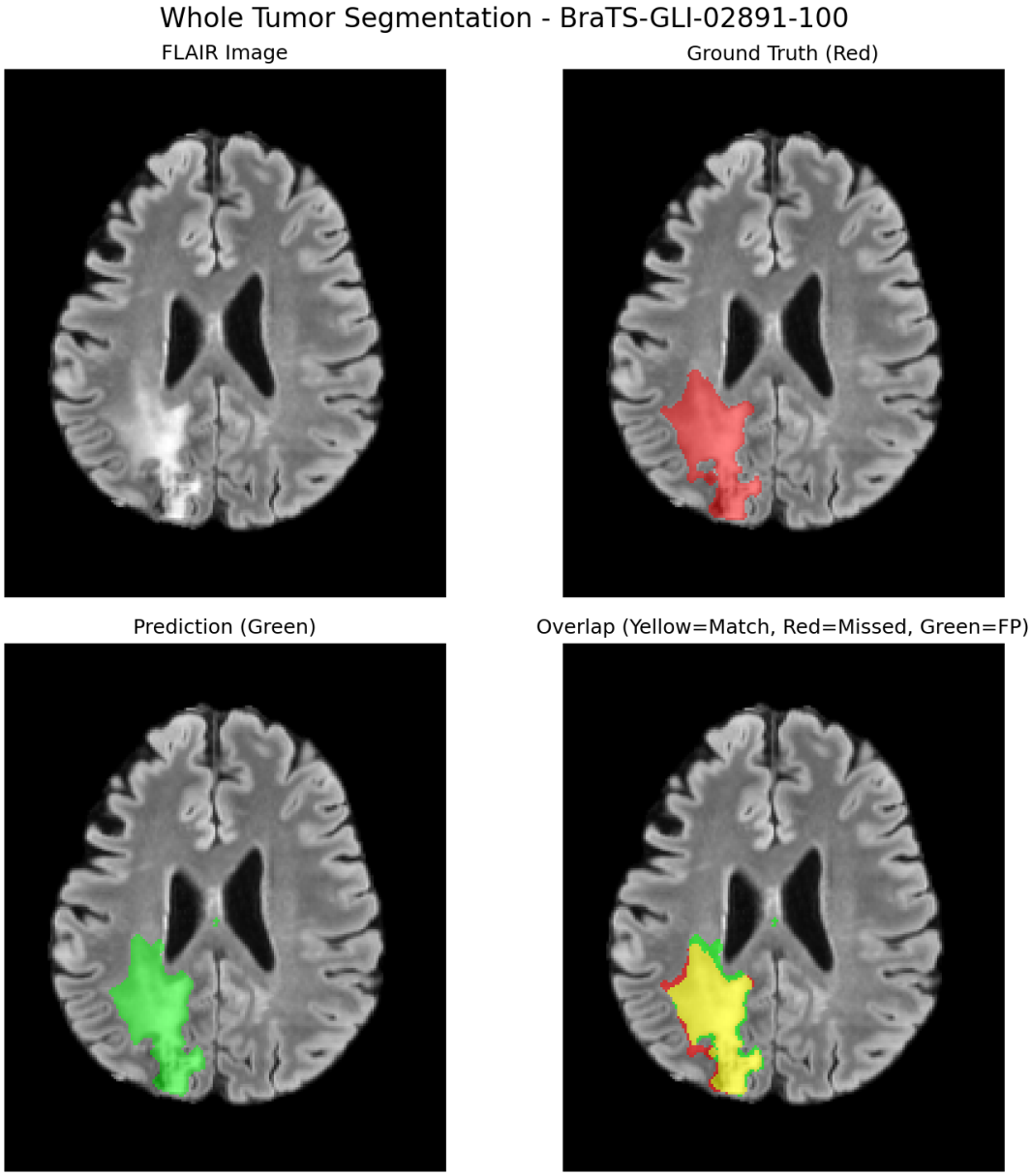

Figure 1: Segmentation visualization for BraTS-GLI-02891-100

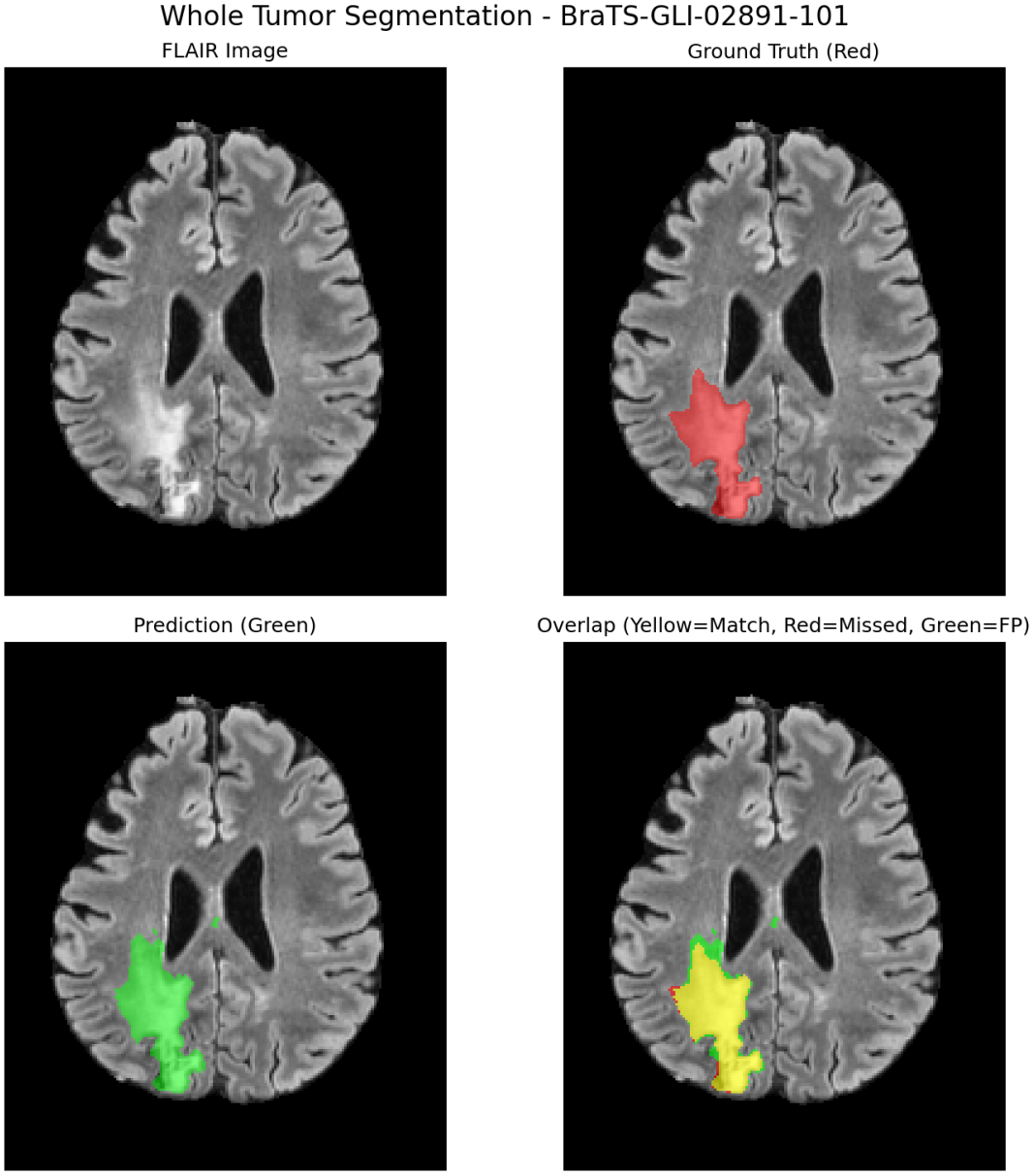

Figure 2: Segmentation visualization for BraTS-GLI-02891-101

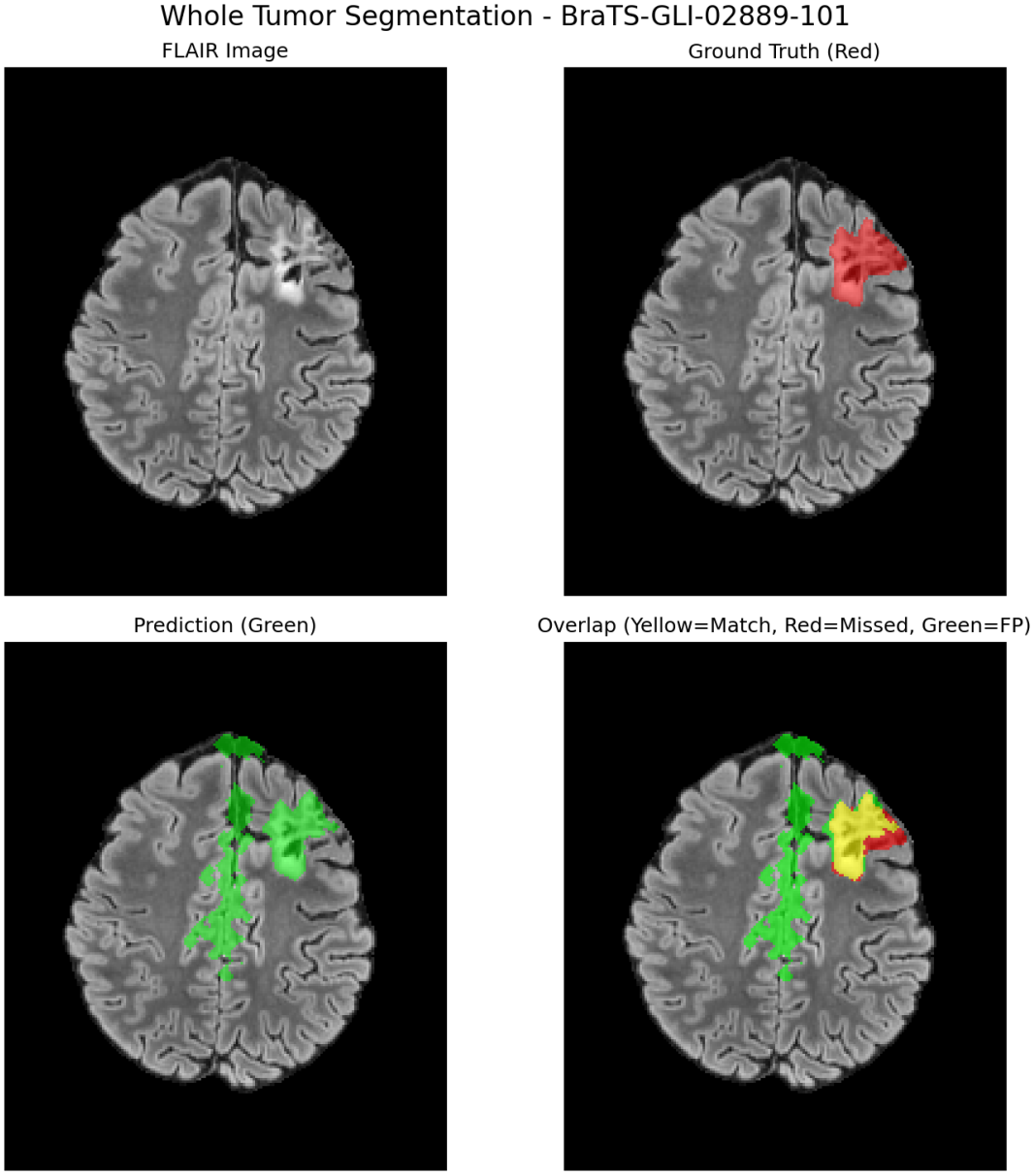

Figure 3: Segmentation visualization for BraTS-GLI-02889-101

6. Quantitative Evaluation

The Dice Similarity Coefficient was used to evaluate segmentation accuracy.

[Dice = 2—P $\cap G|_{\overline{|P|+|G|]}}$

where (P) represents predicted tumor voxels and (G) represents ground truth tumor voxels.

Metric	Value
Total Test Cases	270
Mean Dice Score	0.6783
Median Dice Score	0.7573
Standard Deviation	0.2237

Table 1: Validation Performance Summary

7. Observations

- The model successfully detects major tumor regions.
- Visualization results show strong overlap between predicted and ground truth masks.
- Some scattered false positives occur due to similar intensity patterns.
- Boundary regions of tumors are occasionally missed.

8. Plan for Week 4

The next stage will focus on further evaluation and analysis of the segmentation pipeline.

- Running the full pipeline on additional validation cases
- Computing Dice scores for all patients
- Analyzing prediction errors
- Improving post-processing techniques
- Preparing the final project report

9. Conclusion

Week 3 focused on implementing the complete tumor segmentation pipeline using classical machine learning techniques. The system integrates feature engineering, Random Forest classification, and morphological post-processing to generate tumor segmentation masks. Initial results demonstrate that classical methods can achieve meaningful segmentation performance without requiring deep learning architectures.